

SUPPLEMENT: NON-COMMUNICABLE DISEASE PREVENTION

Personalised risk-stratified approaches in prevention of cancer and other NCDs: The Joint Action PreventNCD project

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Abstract

Aims: Effective non-communicable diseases prevention and risk reduction require a personalised approach tailored to individuals' characteristics. This work package of the Joint Action PreventNCD focuses on prevention at the individual level, providing evidence on the role that genetic susceptibility, as well as behavioural risk factors (such as tobacco use, or physical inactivity) play in prevention, and integrating the outcomes into a holistic approach for the prevention of cancer and other non-communicable diseases. An overview of existing personalised risk-stratified prevention strategies in the area of cancer and major non-communicable diseases will be provided, together with their ethical, legal and social implications. Finally, we will contribute to the development of a health communication framework for the prevention of non-communicable diseases, building on the risk communication strategies used during COVID-19 pandemic period. **Methods:** To achieve the aims of this work package, a combination of different methods will be used, spanning from literature reviews, to public engagement activities, to innovative pilot studies covering different genetic and behavioral risk factors driving cancer and some other non-communicable diseases (namely, genetic susceptibility, tobacco and nicotine consumption, and physical inactivity). **Conclusions:** The interventions and activities in this work package will provide evidence on how the adoption of prevention strategies based on the concept of 'personalisation' and 'risk stratification' can contribute to alleviating the burden of cancer and non-communicable diseases by targeting the right population with the right intervention.

Keywords: JA PreventNCD, personalised prevention, genetic determinants, ELSI, risk communication, tobacco cessation, physical activity

Background

'Risk' is the probability that an event with negative consequences occurs. When talking about cancer and other non-communicable diseases (NCDs), the term

'risk' is most often used to describe the probability of an individual developing a disease, such as cardiovascular diseases or cancer, relapse, experience complications, have a poor quality of life after the diagnosis or treatment and, finally, to die from the disease. This

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likelihood is influenced by a combination of risk factors and health determinants, which might be responsible for different types of cancers and other NCDs. Risk factors are defined as ‘social, economic or biological status, behaviours or environments which are associated with or cause increased susceptibility to a specific disease, ill health, or injury’ [1]. These risk factors operate within a broader framework of health determinants, which are defined as ‘the range of personal, social, economic and environmental factors that determine the healthy life expectancy of individuals and populations’ [1]. These determinants include modifiable factors, such as socioeconomic status, education, and access to healthcare, as well as non-modifiable factors, such as age and genetic attributes. In 1987, an initiative of the European Commission was developed for the first time by the International Agency for Research on Cancer (IARC): the European Code Against Cancer (ECAC). This code lists 12 recommendations on risk factors with high evidence for cancer (but also other NCDs) development, including tobacco, alcohol, physical inactivity, sun exposure, radiation and unhealthy diets [2]. The ECAC provides general recommendations applicable to all individuals regardless of their specific risk profiles. The ‘one-size-fits-all’ approach to prevention tends to treat everyone the same, without accounting for the individual differences in genetic background, lifestyle, and environmental exposures that can produce a unique set of causes for specific diseases [3].

To address these individual differences, enhance the effectiveness of prevention strategies for cancer and NCDs, and ensure their sustainability, tailored approaches have emerged as promising strategies in healthcare. While these approaches are increasingly adopted in curative medicine, their application in prevention lags behind [3].

According to the definition given by the EU-funded project A PeRsOnalized Prevention roadmap for the future HEalThcare (PROPHET), ‘personalised prevention aims to prevent onset, progression and recurrence of diseases through the adoption of targeted interventions that consider the biological information (e.g. genetic and other biomarkers, demographics, health conditions), environmental and behavioural characteristics, the socioeconomic and cultural context of individuals’ [4]. Personalised strategies can, thus, be applied to all the levels of prevention, from reducing the risk of NCD manifestation (primary prevention), to intercepting the disease as early as possible (secondary prevention) in order to manage it optimally with minimal impact on the patient regarding health, quality of life and rehabilitation and reduced risk of

new occurrence (tertiary prevention) and even for relatives in case of identified genetic predisposition. Within the Joint Action Cancer and other NCDs prevention – Action on Health Determinants (JA PreventNCD; see website <https://www.preventnecd.eu/>) [5], the work package ‘Identify individuals at risk – prevent and treat’ focuses on the concepts of individual ‘risk’ and ‘risk stratification’, integrating them into the definition of personalised prevention. Risk stratification systematically categorises individuals based on their specific health risk profiles and preferences. This work will provide theoretical and empirical insights into the use of personalised interventions based on risk stratification, referred to throughout the paper and the work package as ‘personalised risk-stratified prevention’. Given that the activities in this work address mainly cancer, we explicitly refer to ‘cancer and other NCDs’.

While most of the pilot studies focus on one specific risk factor or health determinant, synergies will be created where possible, not only between studies within the work package, but also with other ongoing (or past) studies external to this project.

Objectives

The activities within this work package will support the main aim of the project – and of the Europe’s Beating Cancer Plan (EBCP) – of reducing the burden of cancer and other NCDs mainly at the personal level by producing evidence on the role of genetic determinants in prevention, as well as on risk factors such as tobacco use, unhealthy diets, and physical inactivity, integrating them into a holistic approach for the prevention of cancer and other NCDs. This comprehensive approach to prevention refers to the importance of addressing the diverse and specific characteristics and needs of individuals and patients with cancer and other NCDs, and simultaneously to consider the impacts of interventions and treatments on their physical, emotional, and psychosocial wellbeing. By embracing an individual-centered approach, we aim to reduce the incidence of cancer and other NCDs in Europe. Last but not least, this holistic and individualised approach aims to be equally accessible to all European citizens.

The activities build on the ECAC to steer the implementation of interventions directed to individuals at risk for cancer and other NCDs.

The specific objectives of this work are:

1. To provide a state of the art of existing and emerging personalised risk-stratified prevention approaches in the area of cancer and major

- NCDs; and to investigate the effectiveness of these strategies in identifying individuals at risk and improving health outcomes (Task 1);
2. To deliver ethical, legal and social implications (ELSI) recommendations for personalised and risk-stratified prevention for cancer and other NCDs based on citizen and patient engagement activities (Task 2);
 3. To incorporate lessons learned from European risk communication strategies in the context of health crisis (particularly the COVID-19 pandemic) into effective risk-stratified prevention approaches for cancer and other NCDs (Task 3);
 4. To implement personalised risk-stratified prevention and treatment strategies through pilot studies targeting genetic determinants, tobacco and nicotine cessation approaches, health promotion through physical activity and a holistic approach to individuals and patients with cancer (Task 4).

Methods

To achieve the specific objectives of this work package, different methods will be used. A combination of reviews, public engagement activities, expert consultations and innovative pilot actions will contribute to provide insights on the implementation of personalised risk-stratified approaches in prevention.

Evidence reviews

A first scoping literature review is conducted to provide an overview of the existing and emerging personalised risk-stratified prevention approaches in the field of cancer and major NCDs at the three levels of prevention (primary, secondary, and tertiary). In particular, the review aims to identify approaches targeting the five major risk factors (tobacco use, physical inactivity/sedentary behaviour, unhealthy diet, alcohol use, and genetic predisposition) responsible for NCDs, targeting cancer (all types), cardiovascular diseases, diabetes, chronic respiratory diseases, and mental health conditions. The search specifically targets consensus statements, guidelines, and recommendations from scientific societies, study groups, governmental agencies, and health ministries regarding approaches that categorise the population based on risk levels and propose personalised interventions accordingly.

A desk review is also conducted to examine COVID-19 risk communication strategies across different European countries and the lessons learned from them. While not specific to NCDs, the recent large-scale public health emergency of COVID-19 represents a real-world context in which communication with higher-risk individuals, including people

with chronic conditions, was crucial. For this reason, COVID-19 communication responses provide valuable insights into institutionalised approaches, challenges, and good practices that can inform strategies for NCD prevention and control. The review focuses on official reports, guidelines, white papers, position papers, and policy briefs from international, European, and national (health) organisations, with a particular emphasis on the countries participating in this activity (Germany, Belgium, Norway).

Public engagement activities

Different types of activities will be performed in four European countries (France, Iceland, Norway, Sweden), which will revolve around the questions linked to the goal of risk-stratified prevention: What should determine a risk score (when elaborating individuals' risk profile)? To what extent is it acceptable to discriminate, based on which markers, for which purposes? What is risk stratification useful for? When is stratification justified?

We will investigate opinions, concerns, suggestions from stakeholders on the ethical and legal impact of implementing such strategies within society, building on the findings of the previous EU-funded projects Cancon, iPAAC, TEHDAS and CAN.HEAL [6–9].

Expert consultations

In several activities of the work package, experts in the field of cancer, NCDs and health communication, as well as other stakeholders will be consulted to provide input or feedback on the work done by the partners. For example, in the context of the activities related to lessons learned from risk communication, different consultations will be organised inviting experts who work (or worked) in the area of communication during the pandemic. In a first phase, these consultations (in the form of interviews and focus groups) aim to collect experts' opinions and experiences on what we have learned from the pandemic that can be applied in the context of NCD prevention. In a second phase, workshops will be used to present recommendations elaborated in relation to health communication for NCD prevention.

Innovative pilot actions

Clusters of pilots – organised into and within sub-tasks – covering different risk factors that drive cancer and other NCDs will be conducted with partners from different European countries (see Table I). A first cluster will explore the importance of genetic

Table I. Summary of pilot studies in the work package ‘Identify individuals at risk – prevent and treat’ of the JA PreventNCD.

Theme	Topic (title)	Study design/methods
Genetic determinants	Genetic counselling of familial paediatric-AYA (adolescents and young adults) cancers	Real-world study (survey)
	Integration of genetic determinants into electronic health records	Mapping exercises (literature review, survey), workshop
	Identifying individuals at high risk for cancer and NCDs using polygenic risk scores	Population-based and clinical prospective/retrospective studies
	Towards risk-based breast cancer prevention: lifestyle intervention	Clinical trial and real-world studies
	Preventing second cancers with Dostarlimab immunotherapy (PREDOSTAR)	Clinical trial study
Tobacco and nicotine cessation	Increasing smoking cessation rates by supporting ‘unsuccessful quitters’	Cross-sectional study (literature review, qualitative interviews)
	Piloting the digital tool for individual smoking profile in general practice	Product development and implementation (quantitative analysis, feasibility study)
	Supporting the efficient implementation of early identification and brief intervention related to tobacco and novel nicotine products	Cross-sectional study (literature review, survey, focus group interviews, workshop)
	Developing cessation practices together with tobacco and nicotine users and healthcare services	Cross-sectional study (quantitative and qualitative analysis)
	Free medicines for smoking cessation	Retrospective study (qualitative and quantitative analysis)
	Long-term adult smokers’ barriers to smoking cessation and their perception of cessation services and tobacco control	Cross-sectional study (qualitative interviews)
	Tobacco cessation in group format	Non-experimental descriptive pre-post study (surveys, training workshops)
	Tobacco and nicotine use cessation during pregnancy	Prospective observational study (development of motivational materials and a guide, training, qualitative interviews)
	Effectiveness of in-person, telehealth, and hybrid approaches to smoking cessation in a cancer centre clinic	Retrospective cohort study (quantitative data analysis)
	Promoting physical literacy and health behaviour among Estonian conscripts undergoing military training	Intervention mapping and implementation science (qualitative and quantitative data analysis)
Health promotion: physical activity	Lifestyle intervention for cancer patients: nutrition, physical activity, and quality of life	Feasibility study in a real-life setting (qualitative and quantitative data analysis)
		Descriptive evaluation study
Patients with cancer – a holistic approach to prevention	A holistic intervention model for smoking/nicotine cessation for patients with cancer	
	Long-term outcomes in adolescents and young adults with cancer	Retrospective study based on population-based data
	Replication of a collaborative platform in sexual and reproductive health rehabilitation	Implementation project (scale up an existing digital platform)
	Pilot-Intervention for Informal Cancer Caregivers: PREV-CARE multicentre study	Transfer study, development and testing of a digital app, feasibility and acceptability study
	Shared decision making	Qualitative training intervention
	The Digital Health Centre pilot – a holistic digital cancer rehabilitation programme	Descriptive evaluation study
	Cancer clock – tailored nicotine interventions and digital access	Mixed methods, non-randomised implementation study
	Cancer clock – crossing sectors	Participatory action research design
	Cancer clock – screening	Randomised controlled trial with a mixed-methods evaluation

determinants with different risk levels into the healthcare system to identify individuals with a high likelihood of developing certain illnesses, including cancer (either high penetrant monogenic or low penetrant polygenic predisposition). Even though non-directly actionable, their identification allows us to provide genetic counselling, comprising proposals for tailored interventions on other actionable risk factors, such as lifestyle, to individuals identified as having a genetic risk. A second cluster will promote tobacco and nicotine cessation approaches in various ways and contexts, such as through the analysis of long-term adult smokers’ barriers to smoking cessation, providing free medication as part of cessation

services, or smoking cessation for some vulnerable groups; for example, pregnant women and mental health patients. These pilots will focus on helping healthcare providers offer better support for tobacco and nicotine cessation, with the goal of making these improvements a regular part of the care. A third cluster of pilot studies will focus on the promotion of physical activity among both healthy adults and cancer patients/survivors. A fourth cluster will address several aspects of being diagnosed with cancer, the consequences of treatment, and how optimal patient involvement may reduce the experienced disease burden for the individual. The activities in this cluster will focus not only on the individuals, but also on

the healthcare professionals and healthcare services, and will tackle the full lifecycle perspective (children, adolescents, adults, seniors) using a holistic approach.

A common denominator between the studies will be evaluation of their feasibility and analysis of barriers to their implementation, with a final goal to make recommendations on how to extend the actions at the European level. Two main types of interventions will be used to collect evidence-based qualitative and quantitative data: real-world studies (genetic determinants, tobacco and nicotine cessation, health promotion through physical activity) and clinical trial (for example, immunotherapy for the prevention of second primary cancer in at high-risk individuals).

Expected results

The activities of the work package ‘Identify individuals at risk – prevent and treat’ will contribute to the project by providing quantitative and qualitative insights on personalised prevention strategies based on a risk-stratified approach for the identification of individuals at risk of cancer and other NCDs.

The expected outputs include:

- A report covering the review of the latest evidence for personalised risk-stratified approaches in prevention;
- An ethical framework for personalised risk-stratified prevention for cancer and other NCDs and a public engagement toolkit to foster citizens’ participation in the topic;
- Recommendations on how to build effective risk-adjusted communication strategies;
- Contributions to the EU Best Practice Portal on Public Health [10] and identification of sustainable elements of the pilot actions;
- An EU policy brief on personalised risk-stratified prevention for cancer and other NCDs including recommendations.

Figure 1 summarises the structure of the work package, the methods used for each of the activities and the expected outputs.

Discussion

Despite individuals’ differences in the risks of developing NCDs, most prevention strategies to date have adopted a population-based approach, which builds on a ‘one-size-fits-all’ principle [3]. This principle applies the same guidelines, recommendations and programmes to the entire population, without considering differences in individuals’ risk levels. As

stated in the EBCP, effective cancer prevention and treatment require a personalised approach customised to patients’ characteristics, their environment and the specific nature of the disease [11]. Tailoring approaches to individuals’ specific characteristics and needs can contribute to reducing the impact of NCDs on health services, with better health outcomes compared with the traditional approaches to prevention [12]. Environmental factors are outside the scope of our activities, as they are the focus of other work within the joint action. However, we need to be aware of the potential consequences of individual-level interventions. In the definition provided by the PROPHET consortium, personalised prevention should be ‘timely, effective, and equitable’ [4]. This means that we should ensure that interventions are done on time (to identify the disease before its onset, and after its interception, to ensure accurate treatment); that they are effective; and that they do not contribute to reinforcing socioeconomic inequalities. For example, information given in public health messages might be more easily understood by people with better health literacy [13]. Being identified as part of a higher-risk group – for example in the context of hereditary cancer disclosure – can cause significant psychological distress and anxiety for individuals [14]. Moreover, the information regarding these groups could be misused by health insurances [15]. Genomic medicine has also been criticised for fostering an individualistic approach to healthcare, which could undermine the collective nature of risk pooling that supports social solidarity, increasing the potential for stigmatisation and discrimination [8]. Finally, due to the need for tailored interventions, individualised risk assessments and specialised interventions, personalised prevention strategies might be more costly compared with one-size-fits-all approaches [16]. Our activities will carefully consider these potential harms and aim to investigate further the effects of personalised interventions.

Conclusions

The activities included in the work ‘Identify individuals at risk – prevent and treat’ will contribute to the main aims of the JA PreventNCD by providing a comprehensive overview of existing and emerging personalised risk-stratified prevention approaches in the area of cancer and major NCDs, an ethical framework for personalised prevention and recommendations for effective health communication for NCD prevention. Moreover, evidence will be produced on the integration of genetic risk factors and specific interventions for cancer risk reduction, and the role

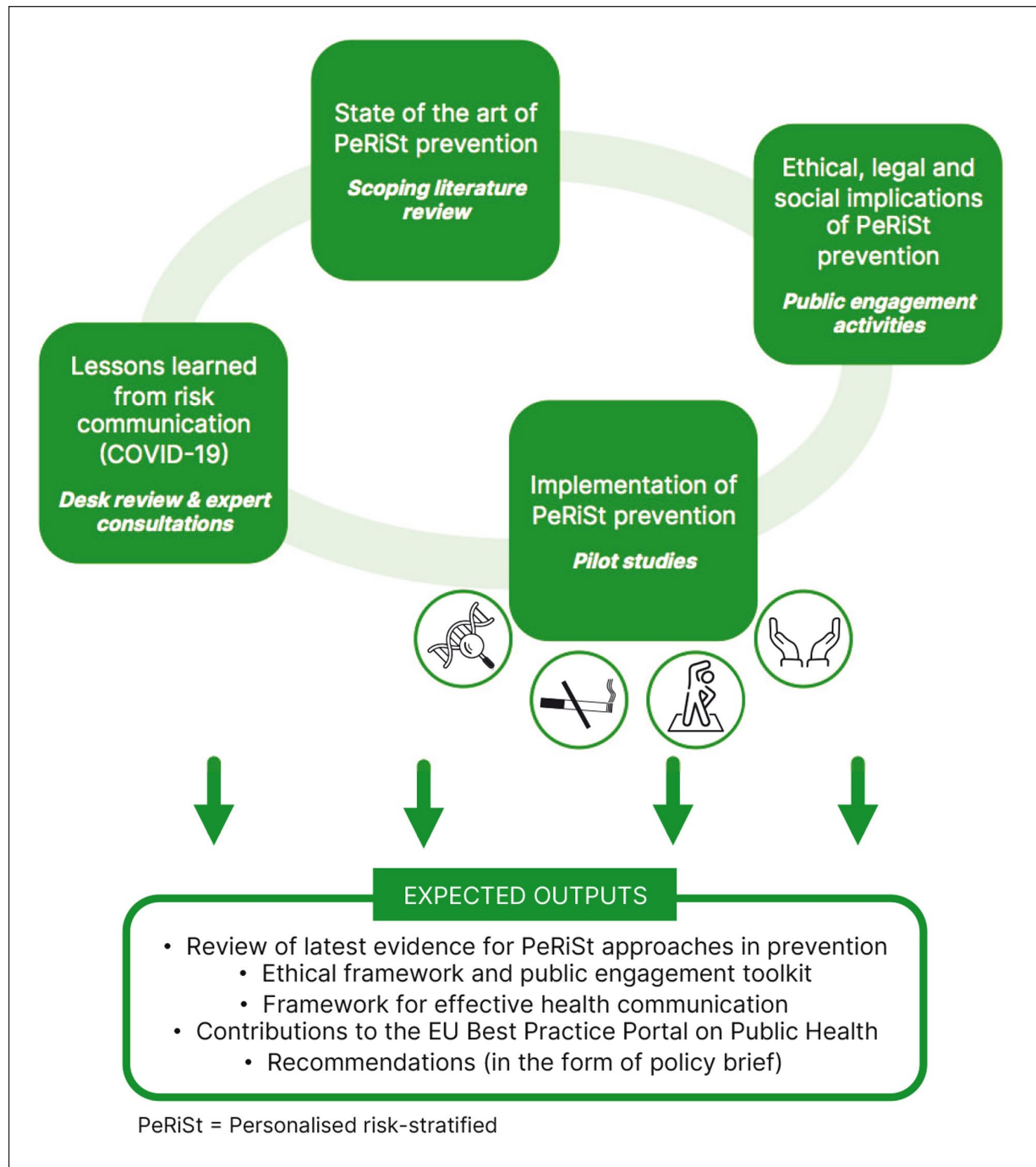


Figure 1. Structure of the work package, methods and expected outputs.

of other personalised interventions covering different risk factors (tobacco use, unhealthy diets and physical inactivity). The outcomes will be integrated into a holistic approach for the prevention of cancer and other NCDs.

Personalised risk-stratified prevention can effectively reduce the burden of cancer and NCDs by integrating genetics, behavioural, and social risk

factors into targeted interventions. Evidence from research, stakeholder engagement, and pilot studies supports shifting from a one-size-fits-all approach to tailored, risk-based prevention/risk reduction strategies; at the same time, ensuring equitable access is crucial to prevent widening health disparities and promote social responsibility in healthcare.

Findings from the activities within this work will inform European policies and best practices, promoting sustainable integration of personalised prevention into public health.

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